# Algebra and Discrete Mathematics (ADM)

## Tutorial 7 Fundamental spaces and decompositions

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# Elementary row operations and row space

**Theorem**

*Elementary row operations do not change the row space of a matrix.*

- It is clear that swapping rows does not affect the row space
- If we multiply a row by a nonzero scalar, we replace a row vector by a nonzero scalar multiple of itself. Every vector in the span of the original rows can still be written as a linear combination of the modified rows, because vector spaces are closed under scalar multiplication.
- If we replace $R_j$ with $R_j + \alpha R_i$, then every vector in the span of the original rows can still be written as a linear combination of the modified rows, because vector spaces are closed under addition and scalar multiplication.

# Elementary row operations may change column spaces

Consider

$$A = \begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$$

$B$ can be obtained from $A$ by the row operation

$$A \xrightarrow{R_2 \to -2R_1 + R_2} B$$

- Column space of $A$ consists of scalar multiples of $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
- Column space of $B$ consists of scalar multiples of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

# LU-decomposition – Doolittle decomposition

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 3 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ \ell_{21} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} u_{11} & u_{12} \\ 0 & u_{22} \end{pmatrix}$$

- Row 1

$$u_{11} = 2, \quad u_{12} = -1$$

- Row 2

$$\ell_{21} u_{11} = 3 \Longrightarrow \ell_{21} = \frac{3}{2}$$

$$\ell_{21} u_{12} + u_{22} = 3 \Longrightarrow u_{22} = 3 - \frac{3}{2} \times (-1) = \frac{9}{2}$$

- We have

$$L = \begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 2 & -1 \\ 0 & \frac{9}{2} \end{pmatrix}$$

# Solving linear system

Consider the linear system $A\boldsymbol{x} = \boldsymbol{b}$, where

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 3 \end{pmatrix}, \quad \boldsymbol{b} = \begin{pmatrix} 0 \\ 9 \end{pmatrix}$$

Since $A = LU$, we first solve $L\boldsymbol{y} = \boldsymbol{b}$.

$$\begin{pmatrix} 1 & 0 \\ \frac{3}{2} & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \end{pmatrix} \Longrightarrow y_1 = 0,\ y_2 = 9$$

$U\boldsymbol{x} = \boldsymbol{y}$

$$\begin{pmatrix} 2 & -1 \\ 0 & \frac{9}{2} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 9 \end{pmatrix} \Longrightarrow x_2 = 2,\ x_1 = 1$$

# Eigenvalues and eigenvectors

$$A = \begin{pmatrix} 5 & 4 \\ 8 & 9 \end{pmatrix}$$

The characteristic equation of $A$ is

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 5 & -4 \\ -8 & \lambda - 9 \end{vmatrix} = (\lambda - 5)(\lambda - 9) - 32 = 0$$

Solving for $\lambda$, we obtain

$$\lambda^2 - 14\lambda + 13 = 0 \Longrightarrow (\lambda - 13)(\lambda - 1) = 0 \Longrightarrow \lambda_1 = 1,\ \lambda_2 = 13$$

# Eigenvalues and eigenvectors

$$A = \begin{pmatrix} 5 & 4 \\ 8 & 9 \end{pmatrix}, \quad \lambda_1 = 1,\ \lambda_2 = 13$$

$$\begin{pmatrix} \lambda - 5 & -4 \\ -8 & \lambda - 9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \xrightarrow{\lambda_1 = 1} \begin{pmatrix} -4 & -4 \\ -8 & -8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = t \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

A basis for the eigenspace corresponding to eigenvalue $\lambda_1 = 1$ is

$$\left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}.$$

$$\begin{pmatrix} \lambda - 5 & -4 \\ -8 & \lambda - 9 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \xrightarrow{\lambda_2 = 13} \begin{pmatrix} 8 & -4 \\ -8 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Longrightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = t \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

A basis for the eigenspace corresponding to eigenvalue $\lambda_2 = 13$ is

$$\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix} \right\}.$$

# Quiz two information

- Time: 9:30–11:50
- Location: same room as the lecture
- Toilet breaks are not permitted during the quiz.
- Calculators are not allowed.
- Write your answers on the provided answer sheets. Additional sheets will be supplied upon request.
- Write your full name clearly on every page of your answer sheets.
- Show all computation steps in sufficient detail. Answers without supporting calculations will receive zero points, even if the final answer is correct.

# Quiz two information

- Q1: vector space axioms, nonstandard vector addition and scalar multiplication, additive identity, and additive inverses
- Q2: linear independence, basis and spanning sets, coordinate vectors relative to a basis, and change-of-basis matrices
- Q3: linear transformations and their standard matrices, row space, range, kernel, rank and nullity, characteristic equations, eigenvalues and eigenspaces, and LU decomposition using Doolittle's method

# Quiz Two: absence and make-up policy

- If you cannot attend Quiz 2, you must submit a medical certificate (MC) to the Study Department.
- Make sure your absence is officially **recorded in UIS**.
- Students with a properly documented absence will be eligible to take the make-up quiz.

# Make-up Quiz

- **Make-up Quiz 1**
  - Week 11
  - Time: 9:30–11:50
- **Make-up Quiz 2**
  - Week 12
  - Time: 11:00–13:20